higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE

## MATHEMATICS N5

## 1 APRIL 2019

**This marking guideline consists of 12 pages.**

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**QUESTION 1**

1.1 1.1.1 $\lim\limits_{x\to 0}\dfrac{3^x-2^x}{x^2-x}$

L'hospital's rule since the condition $\dfrac{0}{0}$ holds; thus

$\lim\limits_{x\to 0}\dfrac{3^x\ln 3-2^x\ln 2}{2x-1}$

$\ln 3-\ln 2$

$=0.405$ (2)

1.1.2 $\lim\limits_{x\to 0}\tan x\ln x$

$=\lim\limits_{x\to 0}\dfrac{\ln x}{\dfrac{1}{\tan x}}$

L'hospital's rule since the condition $\dfrac{\infty}{\infty}$ holds; thus

$\lim\limits_{x\to 0}\dfrac{\dfrac{1}{x}}{-cosec^2 x}$

$\lim\limits_{x\to 0}\dfrac{-\sin^2 x}{x}$

L'hospital's rule since the condition $\dfrac{0}{0}$ holds; thus

$=\lim\limits_{x\to 0}\dfrac{-2\sin x\cos x}{1}$

$=0$ (5)

1.2 $f(x)=\dfrac{1}{2-4\cos 2x}$

$2-4\cos 2x=0$

$\cos 2x=\dfrac{1}{2}$

Thus, $f(x)$ is discontinuous at $x=\dfrac{\pi}{6}$ ✓and $\dfrac{5\pi}{6}$✓ (2)

**[9]**

## QUESTION 2

2.1 2.1.1 $f(x)=\frac{1}{\sqrt{9x}}$

$f(x+h)=\frac{1}{3}(x+h)^{\frac{-1}{2}}$

$\text{¿}\frac{1}{3}\left[\frac{x^{\frac{-1}{2}}h^0}{0!}+\frac{\left(\frac{-1}{2}\right)x^{\frac{-3}{2}}h}{1!}+\frac{\left(\frac{3}{4}\right)x^{\frac{-5}{2}}}{2!}+\ldots\ldots\ldots\right]$

$\text{¿}\frac{1}{3\sqrt{x}}-\frac{h}{6\sqrt{x^3}}+\frac{h^2}{8\sqrt{x^5}}+\ldots\ldots\ldots$ (2)

2.1.2 $f(x+h)-f(x)=-\frac{h}{6\sqrt{x^3}}+\frac{h^2}{8\sqrt{x^5}}+\ldots\ldots\ldots$ (1)

2.1.3 $\frac{f(x+h)-f(x)}{h}=-\frac{1}{6\sqrt{x^3}}+\frac{h}{8\sqrt{x^5}}+\ldots\ldots\ldots$ (1)

2.1.4 $\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=-\frac{1}{6\sqrt{x^3}}$ (1)

2.2 $y=\cot^{-1}x$

$\cot y=x$

$-cosec^2 y\frac{dy}{dx}=1$

$\frac{dy}{dx}=-\frac{1}{cosec^2 y}$

$=-\frac{1}{1+\cot^2 y}$

$=-\frac{1}{1+x^2}$ (4)

2.3 2.3.1 $y=\tan\left[\sqrt[3]{3x^2}+\ln(5x^4)\right]$ (3)

$$¿\ sec^2\left[\sqrt[3]{3x^2}+\ln\left(5x^4\right)\right]\times\left(\frac{2\sqrt[3]{3}}{3}\times x^{\frac{-1}{3}}+\frac{1}{5x^4}\times 20x^3\right)$$

2.3.2 $y=\ln^3(\sin x-\cot x)$

$$\frac{dy}{dx}=3\ln^2(\sin x-\cot x)\times\frac{1}{\sin x-\cot x}\times(\cos x+cosec^2 x)$$ (3)

2.4 $y=\left(2x-e^{8x}\right)^{\sin 2x}$

$$\ln y=\sin 2x\ln\left(2x-e^{8x}\right)$$

$$\frac{1}{y}\frac{dy}{dx}=2\cos 2x\times\ln\left(2x-e^{8x}\right)+\sin 2x\times\frac{1}{2x-e^{8x}}\times\left(2-8e^{8x}\right)$$

$$\frac{dy}{dx}=y\left(2\cos 2x\ln\left(2x-e^{8x}\right)+\frac{\sin 2x\left(2-8e^{8x}\right)}{2x-e^{8x}}\right)$$

$$¿\left(2x-e^{8x}\right)^{\sin 2x}\left(2\cos 2x\ln\left(2x-e^{8x}\right)+\frac{\sin 2x\left(2-8e^{8x}\right)}{2x-e^{8x}}\right)$$ (4)

2.5 2.5.1 $2\sin x\cos y=1$

$$2\left[\cos x\cos y+\sin x(-\sin y)\frac{dy}{dx}\right]=0\ 2\cos x\cos y-2\sin x\sin y\frac{dy}{dx}=0$$

$$2\sin x\sin y\frac{dy}{dx}=2\cos x\cos y$$

$$\frac{dy}{dx}=\frac{\cos x\cos y}{\sin x\sin y}$$ (4)

2.5.2 $\frac{dy}{dx}=1=m$

$y=mx+c$

$y=x+c$

$\frac{\pi}{4}=\frac{\pi}{4}+c$

$c=0$

$y=x$ (3)

**[26]**

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## QUESTION 3

3.1 3.1.1 $f(x)=3x^3-9x^2+5x+2$

$f'(x)=9x^2-18x+5=0$

$(3x-5)(3x-1)=0$

$x=\frac{5}{3} \vee x=\frac{1}{3}$

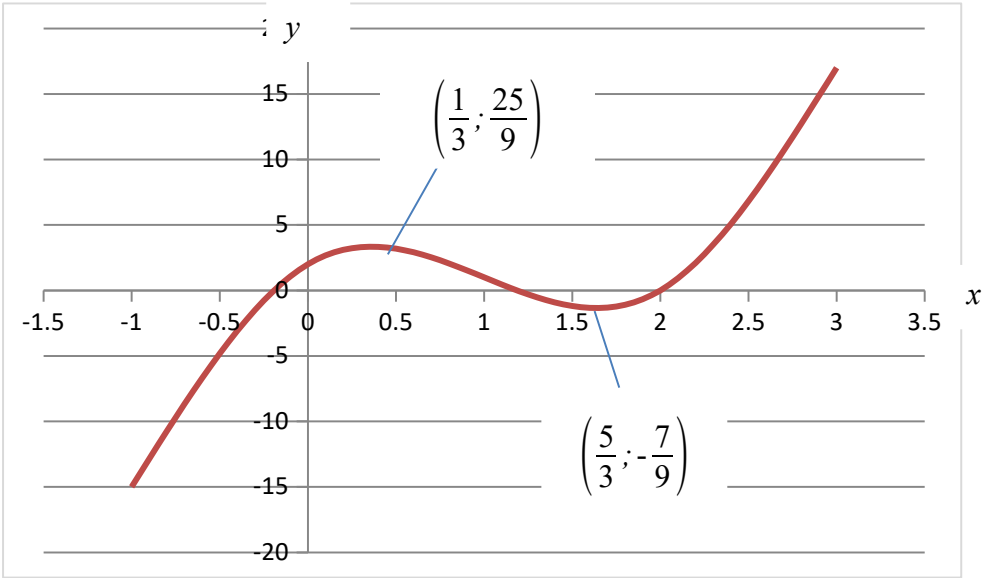
Turning points $\left(\frac{1}{3};\frac{25}{9}\right)$ and $\left(\frac{5}{3};-\frac{7}{9}\right)$ ✓ (2)

3.1.2

| $x$ | -1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|
| $y$ | -15 | 2 | 1 | 0 | 17 |

= 1 mark for any 3 correct answers (2)

3.1.3




= 1 mark for the shape
= 1 mark for the indication of the turning points on the graph (2)

3.1.4 $Let\ \ x_0 = 1$

$f(1) = 1$

$f'(1) = -4$

$x_1 = 1 - \frac{1}{-4}$

$= 1.25$

$Let\ \ x_1 = 1.25$

$f(1.25) = 0{,}046875$

$f'(1.25) = -3{,}4375$

$x_1 = 1.25 - \frac{0{,}046875}{-3{,}4375}$

$= 1{,}263636364$

$\approx 1.264$

$2x + y = 800$ (4)

3.2 $s(t) = e^{t^2} + t^2$

$s'(t) = 2te^{t^2} + 2t$

$s''(t) = 2e^{t^2} + 4t^2e^{t^2} + 2$

$s''(1) = 2e^1 + 4(1)^2e^1 + 2$

$s''(1) = acceleration = 18.31\,m^2/s$ (4)

**[14]**

**QUESTION 4**

4.1 4.1.1 $$\int e^x sec^2\left(e^x\right)\left[1+\tan\left(e^x\right)\right]^{-3}\ dx$$

---

$$let\ \ u=1+\tan\left(e^x\right)$$
$$du=sec^2\left(e^x\right)\times e^x\,dx$$

---

$$¿\int u^{-3}\,du$$

$$¿-\frac{1}{2}u^{-2}+C$$
$$¿-\frac{1}{2}\left[1+\tan\left(e^x\right)\right]^{-2}+C$$ (2)

4.1.2 $$\int \frac{cosec\,x\cot x}{2-cosec\,x}\ dx$$

---

$$let\ \ u=2-cosec\,x$$
$$du=cosec\,x\cot x\,dx$$

---

$$¿\int \frac{1}{u}\,du$$

$$¿\ln u+C$$
$$¿\ln\left(2-cosec\,x\right)+C$$ (2)

4.1.3 $$\int \frac{7}{1+5x^2}dx$$

$$¿\frac{7}{5}\int \frac{1}{\frac{1}{5}+x^2}dx$$

$$¿\frac{7\sqrt{5}}{5}\tan^{-1}\left(\sqrt{5}\right)+C$$ (3)

4.1.4 $\int \tan^3 x\,dx$

$\therefore \int \tan^2 x \tan x\,dx$

$\therefore \int \left(sec^2 x - 1\right)\tan x\,dx$

$\int sec^2 x \tan x\,dx - \int \tan x\,dx$

---

$let \quad u = \tan x$

$du = sec^2 x\,dx$

---

$\therefore \int u\,du - \int \tan x\,dx$

$\therefore \frac{1}{2}\tan^2 x - \ln\left(sec\,x\right) + C$

OR

$\int sec^2 x \tan x\,dx - \int \tan x\,dx$

$\int sec\,x\,sec\,x \tan x\,dx - \int \tan x\,dx$

---

$let \quad u = sec\,x$

$du = sec\,x \tan x\,dx$

---

$\therefore \int u\,du - \int \tan x\,dx$

$\therefore \frac{1}{2}sec^2 x - \ln\left(sec\,x\right) + C$ (5)

4.1.5 $\int \sin(\ln x)\,dx$

---

$let \quad u=\sin(\ln x) \qquad dv=dx$

$du=\cos(\ln x)\times\frac{1}{x}\,dx \qquad v=x$

---

$$\int u\,dv=uv-\int v\,du$$

$$\int \sin(\ln x)\,dx=x\sin(\ln x)-\int x\times\cos(\ln x)\times\frac{1}{x}\,dx$$

$$=x\sin(\ln x)-\int \cos(\ln x)\,dx$$

$$=x\sin(\ln x)-\left[x\cos(\ln x)-\int x\times\frac{1}{x}\times\sin(\ln x)\,dx\right]$$

$$=x\sin(\ln x)-x\cos(\ln x)-\int x\sin(\ln x)\,dx$$

$$¿\ \frac{x\sin(\ln x)-x\cos(\ln x)}{2}$$

(5)

4.2 $\int \frac{9-9x}{2x^2+7x-4}\,dx$

$$\frac{9-9x}{(x+4)(2x-1)}=\frac{A}{x+4}+\frac{B}{2x-1}$$

$$9-9x=A(2x-1)+B(x+4)$$

$let \quad x=-4, \quad then \quad A=-5$

$$x=\frac{1}{2}, \quad then \quad B=1$$

$$=-5\int \frac{1}{x+4}\,dx+\int \frac{1}{2x-1}\,dx$$

$$=-5\ln(x+4)+\ln(2x-1)+C$$

(5)
**[22]**

## QUESTION 5

5.1
$$\int_{\pi}^{2\pi} (1-3x)\sin\left(\frac{1}{2}x\right)dx$$

---

$$let \quad u = 1-3x \qquad dv = \sin\left(\frac{1}{2}x\right)dx$$
$$du = -3\,dx \quad v = -2\cos\left(\frac{1}{2}x\right)$$

---

$$¿\left[-2\cos\left(\frac{1}{2}x\right)(1-3x)\right]_{\pi}^{2\pi} - 6\int_{\pi}^{2\pi}\cos\left(\frac{1}{2}x\right)dx$$

$$¿\left[-2\cos\left(\frac{1}{2}x\right)(1-3x)\right]_{\pi}^{2\pi} - 12\left[\sin\left(\frac{1}{2}x\right)\right]_{\pi}^{2\pi}$$

$$= 14 - 12\pi$$

$$= -23.699 \qquad (5)$$

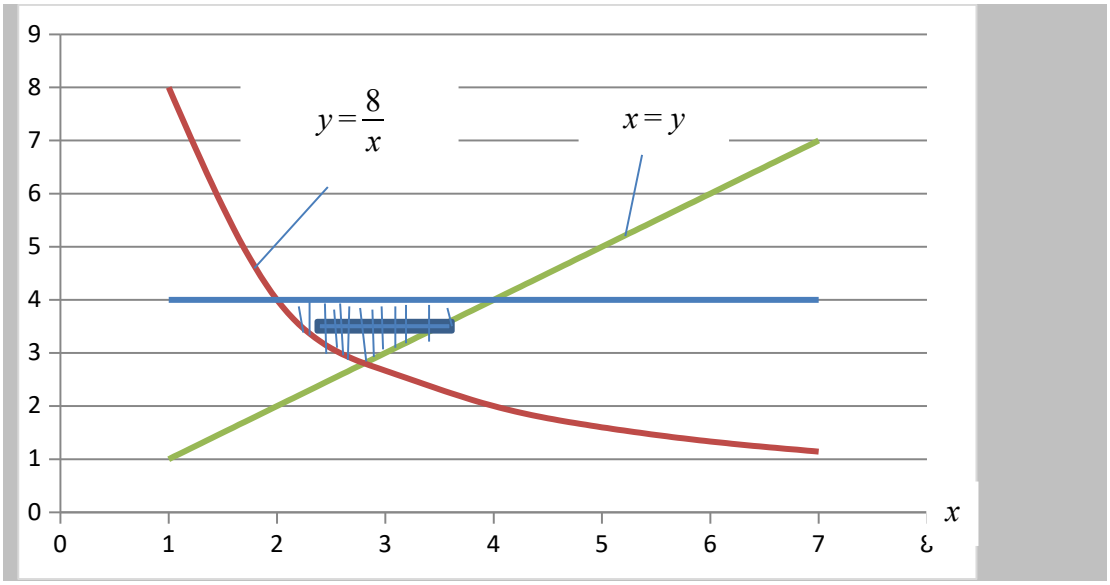
5.2 5.2.1
$$x = \frac{8}{x}$$
$$x^2 = 8$$
$$x = \pm 2\sqrt{2}$$

Thus; coordinates of point of intersection are $(2\sqrt{2}; 2\sqrt{2})$ and $(-2\sqrt{2}; -2\sqrt{2})$ (2)

5.2.2




(2)




= 1 mark for indication of enclosed area
= 1 mark for indication of the vertical or horizontal strip

5.2.3 $A_y=\int_a^b (x_1-x_2)\,dy$

$\int_{\sqrt{8}}^{4}\left(y-\frac{8}{y}\right)dy$

$\left[\frac{y^2}{2}-8\ln y\right]_{\sqrt{8}}^{4}$

$=1.227\,units^2$ (3)

5.2.4 $V_y=\int_a^b (x^2_1-x^2_2)\,dy$

$\int_{\sqrt{8}}^{4}\left(y^2-\frac{64}{y^2}\right)dy$

$\left[\frac{y^3}{3}+\frac{64}{y}\right]_{\sqrt{8}}^{4}$

$=7.163\,unit^3$ (4)

5.3 $m=\sigma A$

$dm=\sigma dA$

(5)

$$dI_z = r^2$$

$$\therefore x^2 \sigma dm$$

$$\therefore x^2 \sigma 2\pi x\, dx$$

$$= 2\pi\sigma \int_0^r x^3\, dx$$

$$= 2\pi\sigma \left[\frac{1}{4}x^4\right]_0^r$$

$$= 2\pi\sigma \left[\frac{r^4}{4} - 0\right]$$

$$\therefore \frac{1}{2}\pi\sigma r^4 \text{ units}^4$$

**[21]**

## QUESTION 6

6.1 $\frac{dy}{dx}\tan x = a + y$

$\frac{dy}{dx} = \frac{a+y}{\tan x}$

$\frac{dy}{a+y} = \cot x\, dx$

$\ln(a+y) = \ln(\sin x) + c$ (3)

6.2 $\frac{d^2y}{dx^2} = x^4 - x^2 + 1$; if $\frac{dy}{dx} = 1,\ x = 1$ and $y$ - 2

$\frac{dy}{dx} = \frac{1}{5}x^5 - \frac{1}{3}x^3 + x + A$

$A = \frac{2}{15}$

$\frac{dy}{dx} = \frac{1}{5}x^5 - \frac{1}{3}x^3 + x + \frac{2}{15}$

$y = \frac{1}{30}x^6 - \frac{1}{12}x^4 + \frac{1}{2}x^2 + \frac{2}{15}x + B$

$B = \frac{17}{12}$

$y = \frac{1}{30}x^6 - \frac{1}{12}x^4 + \frac{1}{2}x^2 + \frac{2}{15}x + \frac{17}{12}$ (5)

**[8]**

**TOTAL: 100**

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